

SIOUX FALLS CLIMATE, SD, USA

WMO: 711680

Lat: 43.7336N

Lon: 96.6233W

Elev: 482

StdP: 95.67

Time Zone: -6.00 (NAC)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-23.9	-21.6	-26.8	0.3	-23.7	-24.6	0.4	-21.6	11.8	-4.9	10.9	-5.9	3.3	280	0.571

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.6	31.3	23.0	29.8	22.1	28.3	21.4	24.8	29.3	23.7	28.2	22.6	27.0	4.8	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.5	19.4	28.0	22.2	17.9	27.0	21.1	16.7	25.7	78.3	29.5	73.5	28.0	69.2	27.1	28.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.0	9.3	8.4	-28.4	33.3	4.2	2.0	-31.4	34.7	-33.8	35.9	-36.2	37.0	-39.2	38.5		
(5)			-29.1	26.5	4.1	0.9	-32.1	27.1	-34.5	27.7	-36.9	28.2	-39.9	28.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	7.1	-9.2	-8.1	0.1	7.0	14.2	20.3	22.1	20.3	16.2	8.0	0.9	-7.2		
	DBStd	12.44	7.27	6.90	7.28	5.72	5.07	3.58	3.21	3.11	4.34	5.09	6.20	6.43		
	HDD10.0	2467	594	506	314	121	19	0	0	0	5	99	276	532		
	HDD18.3	4411	852	739	564	342	147	20	7	16	91	321	522	791		
	CDD10.0	1426	0	0	8	31	149	311	374	319	192	38	4	0		
	CDD18.3	327	0	0	1	1	18	80	123	77	27	1	0	0		
	CDH23.3	2702	0	0	2	24	203	654	1032	572	204	12	0	0		
	CDH26.7	733	0	0	0	5	54	193	312	131	37	1	0	0		
(14) Wind	WSAvg	4.0	4.0	4.2	4.6	5.0	4.2	3.8	3.2	3.0	3.6	4.1	4.2	4.0		
(15) Precipitation	PrecAvg	695	14	20	37	73	94	112	84	84	69	57	31	20		
	PrecMax	943	35	57	104	172	202	327	210	199	186	156	109	53		
	PrecMin	444	1	4	4	12	25	25	11	27	6	8	1	3		
	PrecStd	126	8	12	25	36	47	56	55	47	46	43	30	14		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	6.3	12.9	21.8	27.0	30.9	33.3	33.7	31.7	29.8	24.9	18.3	9.2		
		MCWB	2.9	8.8	14.8	15.5	17.5	22.4	24.2	23.9	21.9	16.1	12.5	6.6		
	2%	DB	3.7	6.0	17.3	21.9	27.9	30.8	31.6	29.6	27.8	21.5	15.1	5.4		
		MCWB	1.6	3.6	11.0	13.2	18.0	21.4	24.5	23.5	20.6	14.1	10.2	2.5		
	5%	DB	1.9	3.4	14.0	18.9	25.6	28.8	30.0	28.1	25.7	18.9	12.0	3.0		
		MCWB	0.7	1.7	9.9	11.4	16.9	20.3	23.5	22.4	19.6	12.8	7.7	1.6		
	10%	DB	0.6	1.4	10.7	16.1	22.9	27.1	28.3	26.6	23.9	16.3	9.7	1.3		
		MCWB	-0.3	0.2	7.3	9.8	15.4	19.7	22.3	21.4	18.5	11.3	6.9	0.3		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.5	9.3	15.9	16.1	21.2	24.9	26.8	25.7	23.4	17.3	13.2	6.9		
		MCDB	5.9	13.0	19.6	23.7	26.1	31.0	31.5	30.0	27.8	22.5	17.0	8.6		
	2%	WB	1.9	4.1	12.3	14.6	19.4	23.2	25.3	24.3	21.9	15.1	10.5	3.0		
		MCDB	3.3	5.7	15.7	20.8	25.5	28.7	29.5	28.2	26.2	19.9	14.3	4.5		
	5%	WB	0.8	1.8	9.8	12.7	18.0	22.0	24.2	23.1	20.8	13.6	8.6	1.7		
		MCDB	1.8	3.3	13.9	17.5	24.0	27.2	29.0	26.7	24.2	18.1	11.6	3.0		
	10%	WB	-0.1	0.3	7.3	10.8	16.6	20.7	23.0	21.9	19.5	11.7	6.7	0.3		
		MCDB	0.7	1.3	11.1	14.9	21.2	25.4	27.5	25.7	22.6	15.5	9.6	1.3		
(35) Mean Daily Temperature Range	MDBR		8.0	8.1	8.7	10.1	10.2	9.3	9.6	9.6	10.4	9.5	8.8	7.7		
	5% DB	MCDBR	9.1	10.2	13.6	15.1	13.7	11.8	10.4	11.0	12.1	13.6	12.2	8.9		
		MCWBR	7.8	7.6	8.3	7.9	6.1	5.4	4.9	5.8	5.9	7.5	7.5	7.0		
	5% WB	MCDBR	8.4	9.7	12.6	12.9	11.4	10.3	9.8	9.9	11.4	11.1	11.1	8.4		
		MCWBR	7.2	7.4	7.9	7.7	6.0	5.9	5.2	5.9	6.1	7.0	7.3	6.8		
(40) Clear-Sky Solar Irradiance	taub		0.269	0.282	0.317	0.368	0.384	0.394	0.401	0.403	0.355	0.325	0.292	0.272		
	taud		2.431	2.416	2.409	2.291	2.323	2.351	2.332	2.299	2.412	2.472	2.482	2.439		
	Ebn at Noon		873	920	925	895	887	876	866	850	871	857	838	832		
	Edh at Noon		73	88	102	125	125	122	123	123	101	83	69	66		
(44) All-Sky Solar Radiation	RadAvg		1.90	2.95	3.85	4.86	5.59	6.26	6.66	5.50	4.38	2.96	2.03	1.51		
	RadStd		0.17	0.23	0.31	0.45	0.41	0.41	0.24	0.42	0.33	0.35	0.14	0.12		

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (38 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page